OncoEvidence: when does a knowledge graph earn its place in oncology drug repurposing?

A mechanism-first, honestly-evaluated evidence-triage pipeline

OncoEvidence

Research preprint — hypothesis-generating, not medical advice

Abstract

Knowledge-graph (KG) neural networks are widely promoted for drug repurposing, usually on the claim that the graph improves link prediction. We test that claim honestly on PrimeKG oncology indications and find it does *not* hold: a properly tuned tabular model (XGBoost) on the identical node features matches or beats a heterogeneous graph neural network (GNN) in every evaluation regime, including inductive cold-disease and cold-drug splits. Rather than discard the graph, we move its job to where a tabular model cannot follow — producing a *traceable multi-hop mechanism-of-action (MOA) path* (drug → target → (PPI/pathway) → cancer gene → cancer) — and we add a literature-grounded LLM layer that checks whether each proposed path is supported by the published record. We evaluate this pipeline instead of demonstrating it. Mechanism structure separates true indications from random pairs at AUROC 0.879 and agrees with curated DrugMechDB mechanisms at 0.802; under deliberately hard negatives the signal is strong against random and oncology-drug controls (0.887/0.870) but only modest (0.609) against shared-target negatives, which we report rather than hide. A prospective time-split shows the graph *enriches a small, top-priority tail*: trained only on pre-2005 structure it surfaces 11× as many future indications in the top 50 as a structure-blind control (recovering post-cutoff approvals such as romidepsin and pralatrexate for cutaneous T-cell lymphoma and nilotinib for CML), although it is not better on median rank. A counterfactual test confirms the mechanism head is faithful: deleting the single curated mechanism edge collapses the prediction far more than deleting a random edge ($p \approx 3.6 \times 10^{-34}$). Going further, a counterfactual stress test shows the prediction is *causal* (target-edge ablation AUROC 0.878), *specific* (a wrong target is rejected 92.6% of the time), and literature-supported over decoys, and independent DepMap/GTEx functional-genomics data corroborate that the bridge genes are real cancer dependencies (AUROC 0.607, $p = 4.6 \times 10^{-6}$). Finally, a popularity-deconfounded ClinicalTrials.gov check is reported as a fully characterized negative. The contribution is not a new model but an evidence-triage pipeline — and an honest account of *when graph structure is actually necessary* — that tests whether a proposed mechanism is causally important, specific, and supported.

1 Introduction

Drug repurposing — finding new oncology uses for existing drugs — is attractive because the safety profile is known and development is cheaper than *de novo* discovery. Biomedical knowledge graphs such as PrimeKG [1] encode drugs, diseases, proteins, and pathways as a heterogeneous graph, and a large literature applies graph neural networks (GNNs) to predict drug–disease links over them [2, 4]. The usual headline is that the graph improves prediction.

We argue that this headline is often an artifact of a missing baseline. When the same node features that feed the GNN are handed to a *tuned* gradient-boosted tree, the tabular model is an extremely strong ranker, and the relational structure adds little to the link task. Reporting this negative result is the honest starting point of our work, and it reframes the question from "does the graph win?" to "*what is the graph uniquely good for?*"

Our answer: the graph's irreplaceable output is not a score but a *mechanism*. A tabular model can rank a (drug, cancer) pair; it cannot tell you *why*. The KG can produce a short, readable, biologically-typed chain from the drug, through its protein target and interaction/pathway neighborhood, to a cancer-associated gene and the cancer. We make that chain the deliverable, filter out phenotype/symptom coincidences, and add an LLM evidence layer that grounds each chain in retrieved literature. We then *evaluate* the whole pipeline against curated mechanisms, hard negatives, a prospective time-split, and a counterfactual faithfulness test.

Contributions.

1. **An honest "is the graph necessary?" audit.** With a tuned XGBoost on identical features (plus topology-removal and memorization-control ablations), the GNN does not beat the tabular baseline on link ranking in any regime (Sec. 3).
2. **Mechanism-path extraction as the graph's real job**, prioritizing MOA relations over phenotype/symptom overlap, with a quantitative true-vs-random separation and an *honest hard-negative stress test* (Sec. 4).
3. **A literature-grounded LLM evidence layer** that reads abstracts (not just titles) and is more precise than a lexical baseline against curated mechanisms (Sec. 5).
4. **Prospective and faithfulness audits**: top-tail enrichment of future indications under a time-split (Sec. 6) and a counterfactual edge-deletion test of mechanism faithfulness (Sec. 7).
5. **A counterfactual evidence-triage layer** (Sec. 8) that tests whether the prediction is *causal* (target-edge ablation), *specific* (wrong-target rejection), and *supported* (decoy-path verification), with independent DepMap/GTEx corroboration and an evidence-weighted retrieval layer.
6. **A popularity-deconfounded orthogonal check** against ClinicalTrials.gov, reported transparently as a negative (Sec. 9).

We do not claim to invent KG- or LLM-based repurposing; close prior work exists [2, 3]. The contribution is the honest evaluation and the mechanism-first design that follows from it.

2 Data and models

We build a heterogeneous graph from PrimeKG [1] with node types for drugs, diseases, genes/proteins, pathways, and biological processes, and the corresponding typed relations (indication, drug–protein, disease–protein, protein–protein interaction, pathway membership). Node features are sentence-embedding encodings of node names/descriptions (with a hashing fallback for offline use). Oncology diseases are flagged via an ontology filter. The target task is the drug–disease `indication` relation.

We compare four models under a shared evaluation harness: (i) a 2-layer heterogeneous GNN with an InfoNCE-style link decoder; (ii) a tuned XGBoost on concatenated endpoint features; (iii) a feature MLP; and (iv) a DistMult KG embedding as a memorization control. We evaluate three

Table 1: Link-prediction benchmark (mean±sd over 5 seeds). The tuned tabular model matches or beats the GNN in every regime; the graph does not earn its place on link ranking alone.

Regime	Model	AUROC	AUPRC	F1
Transductive	XGBoost	**0.988±0.002**	**0.988±0.002**	**0.952±0.003**
	GNN	0.977±0.003	0.972±0.005	0.937±0.005
Cold-disease (onc.)	XGBoost	**0.963±0.007**	**0.954±0.013**	**0.919±0.007**
	GNN	0.882±0.050	0.828±0.052	0.843±0.040
Cold-drug	XGBoost	**0.958±0.007**	**0.957±0.007**	0.906±0.010
	GNN	0.956±0.006	0.943±0.008	**0.933±0.008**

regimes: *transductive*, *inductive cold-disease* (held-out oncology diseases never seen in training), and *inductive cold-drug*. Negatives are sampled with degree awareness and all splits are leakage-checked.

3 The link task does not need the graph

Table 1 reports the four-model comparison over five seeds. In the transductive regime the tuned XGBoost leads (AUROC 0.988 vs the GNN's 0.977). The gap matters most in the inductive cold-disease oncology regime — the setting closest to real repurposing, where the disease is unseen — where XGBoost reaches 0.963 against the GNN's 0.882. The GNN never wins. A topology ablation makes the reason concrete: shuffling the edges barely changes GNN performance, so it is leaning on node features rather than relational structure for this task.

4 The graph's real job: mechanism paths

We connect a drug to a disease through *mechanism* relations only, in three templates of decreasing specificity:

$$\begin{aligned}
\text{direct target:}\quad & \text{drug} \xrightarrow{\text{targets}} P \xleftarrow{\text{assoc}} \text{disease} \\
\text{PPI bridge:}\quad & \text{drug} \xrightarrow{\text{targets}} P_1 \xrightarrow{\text{PPI}} P_2 \xleftarrow{\text{assoc}} \text{disease} \\
\text{shared pathway:}\quad & \text{drug} \xrightarrow{\text{targets}} P_1 \xrightarrow{\text{in}} \text{pw} \xleftarrow{\text{in}} P_2 \xleftarrow{\text{assoc}} \text{disease}
\end{aligned}$$

Phenotype/symptom relations (a drug whose side effect happens to be a disease symptom) are *excluded* by construction, and promiscuous hub intermediates (plasma carriers such as albumin, generic high-degree proteins) are blocked or down-weighted by an IDF-style promiscuity penalty so the path stays specific. Each pair gets a single mechanism score equal to its strongest path.

Does mechanism structure separate true from random pairs? Over 400 true oncology indications and 400 random drug–cancer pairs, the mechanism score separates the groups at **AUROC 0.879**, with a direct-target rate of 34% for true pairs versus 0.25% for random, and the extracted chains recover textbook MOAs (FLT3, RARA, TOP2A, TYMS).

Honest hard negatives. A single number against random pairs is too easy, so we stress-test against progressively harder negatives (Table 2). The signal stays strong against random and oncology-drug negatives (0.887/0.870) and degrades gracefully against degree-matched negatives (0.742), but against *shared-target* negatives — a drug whose target is itself linked to the cancer —

Table 2: Hard-negative stress test (separation AUROC, true vs. each negative class). Strong vs. random/oncology-drug; modest under the hardest shared-target controls.

Negative class	random	oncology-drug	degree-matched	shared-target
Separation AUROC	0.887	0.870	0.742	**0.609**

it falls to 0.609, near chance. This is the honest boundary of the method: mechanism structure is decisive against generic negatives and only weakly discriminative when the negative already shares the same biology. We foreground this rather than reporting only the favorable number.

Curated agreement. Mapped to HGNC symbols, the extracted mechanism genes agree with expert-curated DrugMechDB mechanisms at **0.802** on covered pairs — the paths are not just internally consistent, they match an external ground truth.

5 Literature-grounded LLM verification

For each candidate the pipeline retrieves Europe PMC records (title *and abstract*) and assembles the mechanism path plus the retrieved abstracts into a retrieval-augmented prompt. An LLM writes a structured evidence dossier (mechanistic rationale / supporting / contradicting / verdict) grounded in the supplied abstracts, and a separate LLM-as-judge pass scores plausibility and evidence strength. Against the curated mechanisms, the LLM's *supported* calls are precise (precision 0.857) versus 0.591 for a lexical baseline, and a stricter MOA rubric with sentence-level grounding rejects coincidental hub paths (e.g. albumin bridges). The LLM verifier grades true pairs *supported/weak* and random pairs *no-path* (11/50 vs 1/50 supported).

6 Prospective audit: top-tail enrichment over time

PrimeKG has no timestamps, so we assign each true indication the year of its earliest Europe PMC co-mention and hold out indications established after a cutoff $T = 2005$ (350 sampled pairs, 341 years resolved; 240 past / 101 future; 3 seeds). Trained only on pre-T structure, the graph model ranks held-out future indications at **AUROC 0.930** (vs sampled negatives), beating a structure-blind control by +0.047.

The aggregate ranking story is deliberately nuanced. When we instead rank every drug against each future cancer using only pre-T structure, the GNN's *median* rank (3,019 of ∼7,956) is *worse* than the control's (1,482), and a paired Wilcoxon test does not favor the GNN on overall rank ($p \approx 1.0$). What the graph does is *enrich the top-priority tail*: it places **11 of 101** future indications in the top 50 (a 17.3× enrichment over a random screen) versus **1** for the control, and 13.9% in the top ∼1% versus 1.0%. The hits are recognizable post-cutoff approvals the model had no indication edge for (Table 3). The honest claim is narrow: the graph does not rank future indications better on average; it concentrates a real signal exactly where a repurposing shortlist is read.

7 Faithfulness: is the named mechanism causal to the prediction?

A mechanism path is only meaningful if the model actually uses it. For each held-out pair whose bridge gene is curated, we delete the single mechanism edge and re-score, and compare against deleting a random edge of the same type. Deleting the mechanism edge collapses the prediction

Table 3: Prospective named cases ($T = 2005$, single seed). A model trained on pre-T structure ranks each drug near the top of ~7,956 candidates for a cancer whose indication is first co-mentioned only after T, far above the structure-blind control.

Drug	**Cancer (future indication)**	**1st lit. yr**	**GNN rank**	**MLP rank**
Trabectedin	ovarian carcinoma (subtypes)	2006–10	**2–5**	3038
Methoxsalen	cutaneous T-cell lymphoma	2006	**10**	2154
Romidepsin	cutaneous T-cell lymphoma	2008–10	**63**	1726
Pralatrexate	cutaneous T-cell lymphoma	2009	**321**	2235
Nilotinib	CML, BCR-ABL1 positive	2006	**342**	1098
Plerixafor	plasma cell myeloma	2007	**428**	1797

Table 4: Counterfactual mechanism stress test. The model relies on the true MOA edge (T1), rejects wrong targets (T2), and is supported over decoy paths (T3).

Counterfactual	**Key statistic**	**Sep. AUROC**
T1 target-edge ablation (causal)	faithful 87.2%, $p \approx 10^{-40}$	0.878
T2 wrong-target substitution (specific)	rejection 92.6%	0.923
T3 decoy-path verifier (supported)	supported 0.80 vs 0.17	—

far more than a random edge (paired Wilcoxon $p \approx 3.6 \times 10^{-34}$; faithful in 83.3% of cases). The mechanism the system names is the mechanism the prediction leans on, not a post-hoc story.

We also test mechanism *recovery*: with the direct drug–gene edge hidden, only the *jointly-supervised* GNN names the curated bridge gene (R@10 0.25), where a degree-trivial baseline and a link-only GNN scored through the same head recover ~0. Because the link-only head was never trained on the mechanism task, this isolates the *objective*, not the architecture: joint mechanism supervision creates a recoverable mechanism signal that a link objective alone does not. We do not claim de novo discovery; this is recovery of *known* mechanism through indirect structure on a modest sample.

8 Counterfactual evidence-triage: causality, specificity, orthogonal support

Faithfulness (Sec. 7) asks whether the model uses the named edge. A stronger question — and the project's central claim — is whether the prediction is made for the *right biological reason*. We probe this with three counterfactuals (Table 4). **(i) Target-edge ablation:** deleting the curated drug–target edge drops the mechanism score far more than deleting a matched random edge of the same drug (contrast 0.168, faithful 87.2%, paired Wilcoxon $p \approx 10^{-40}$, separation AUROC 0.878). **(ii) Wrong-target substitution:** replacing the true target with a degree-matched, disease-unassociated decoy is rejected 92.6% of the time (true-vs-decoy AUROC 0.923). **(iii) Decoy-path verification:** swapping the bridge gene for a decoy and re-running the literature verifier, true bridges are graded *supported* 80% of the time versus 17% for decoys. The prediction is causally tied to, specific to, and literature-supported on the named mechanism.

Independent functional-genomics corroboration. The graph and the literature are not independent of each other. We add a third axis from data the model never saw: DepMap CRISPR (Chronos) gene-effect and GTEx tissue expression. The bridge genes of true indications are stronger dependencies in the matched cancer lineage than those of shared-target negatives (AUROC 0.607, Mann–Whitney $p = 4.6 \times 10^{-6}$) and more expressed in the matched tissue (AUROC 0.618), with no reference to PrimeKG or text. Folding functional features into a specificity classifier lifts the hardest separation (true vs shared-target) from the path-only 0.609 to 0.751; the honest control is that a gradient-boosted model on structure-only features already reaches 0.731, so the functional data add a clean increment (+0.054 on a linear model) on top of structure rather than the whole jump. The contribution is orthogonal evidence that the bridge genes are real cancer dependencies.

Evidence weighting makes retrieval load-bearing. Re-weighting each path by Europe PMC evidence (co-mention strength, recency, bridge-gene specificity, minus a contradiction penalty) leaves the saturated true-vs-random separation unchanged (0.886 → 0.890) but sharpens the hard case from 0.618 to 0.742, demoting coincidental hub paths (e.g. an Insulin→RB1→GIST chain with structure score 3.08 collapses to 0 because the drug is never co-mentioned with the cancer). The retrieval layer is thus load-bearing exactly where structure alone fails. For decision-readiness we add split-conformal calibration (empirical coverage 90.3% at a 90% target, abstaining on 96% of negatives), bootstrap 95% CIs on the headline separations, a multi-cutoff temporal trend (T=2000/2005/2010, graph gain $+0.065/+0.037/+0.029$), a mechanism-novelty triage (∼83% of the shortlist are non-textbook MOAs), and a timestamped, hash-committed registry of 60 novel predictions for prospective falsification.

9 A popularity-deconfounded orthogonal check (an honest negative)

Do the model's novel predictions show up as real human trials? A naive check against ClinicalTrials.gov is badly confounded: some drugs are trialed against every cancer and some cancers against every drug. We fit an additive two-way model $\text{score}(d, c) = \mu + \alpha_d + \beta_c + \varepsilon_{dc}$, remove the drug effect α_d and cancer effect β_c, and test whether the residual interaction still predicts a trial. The within-drug signal looks strong (0.821), but once the cancer effect is also removed the interaction AUROC is 0.475 ($p = 0.85$) — no signal. The apparent effect was cancer popularity, not pairwise specificity. We report this as a fully characterized negative; it stands in deliberate contrast to the prospective result, where specific signal *does* appear.

10 Discussion and limitations

The strongest, most defensible framing of this project is not "our GNN beats XGBoost" (it does not), but: *a tuned tabular model is a strong ranker, the graph earns its place by producing testable mechanism paths, and an LLM/retrieval layer checks whether those paths are supported by the literature.* Every claim is paired with the test that could have falsified it.

Limitations are real and stated throughout. The prospective year proxy is a literature co-mention, not a regulatory date. The faithfulness and recovery tests concern *known* mechanism recovered through indirect structure, not de novo discovery, on modest samples. The mechanism signal is only weakly discriminative against shared-target negatives. The ClinicalTrials.gov check is a negative once popularity is removed. And the candidate shortlist is hypothesis-generating, not a vetted clinical recommendation. We view these not as caveats to bury but as the substance of an honest evaluation.

References

[1] Chandak, Huang, Zitnik. Building a knowledge graph to enable precision medicine (PrimeKG). *Scientific Data*, 2023.

[2] Huang, Chandak, *et al.*, Zitnik. A foundation model for clinician-centered drug repurposing (TxGNN). *Nature Medicine*, 2024.

[3] KGML-xDTD: a knowledge graph-based machine learning framework for drug treatment prediction and mechanism description. arXiv:2212.01384.

[4] Zitnik, Agrawal, Leskovec. Modeling polypharmacy side effects with graph convolutional networks (Decagon). *Bioinformatics*, 2018.